



Problem Solving

Try This!

I completed an obstacle course in $6\frac{2}{3}$ min. It took Greg $1\frac{1}{2}$ times as long to complete it.

How much faster was Gary than Greg in minutes?

Solution:

Step 1: Write a number sentence.

$$6\frac{2}{3} \times 1\frac{1}{2} - 6\frac{2}{3} = \boxed{x}$$

Step 2: Evaluate.

$$\begin{aligned} & 6\frac{2}{3} \times 1\frac{1}{2} - 6\frac{2}{3} \\ &= \frac{20}{3} \times \frac{3}{2} - 6\frac{2}{3} \\ &= \frac{10}{1} - 6\frac{2}{3} \\ &= \boxed{3\frac{1}{3}} \end{aligned}$$

Step 3: Write a concluding sentence.

Gary was $\boxed{3:30}$ min faster than Greg.



- ① Refer to the question above. It took George $\frac{3}{4}$ of the time Gary needed to complete the obstacle course. How much faster was George than Gary in minutes?

$$6\frac{2}{3} \times \frac{3}{4} = \frac{20}{3} \times \frac{3}{4} = 5$$

$$\frac{20}{3} - \frac{5}{1} = \frac{5}{3}$$

George was $\boxed{1\frac{2}{3}}$ min faster than Gary.

- ② Milo observed that a Ferris wheel revolves $4\frac{4}{5}$ times in $2\frac{1}{4}$ minutes. How many times does the Ferris wheel revolve in 10 minutes?

$$4\frac{4}{5} \div 2\frac{1}{4} = \frac{248}{5} \times \frac{4}{93} = \frac{32}{15} = \frac{32}{15} \times \frac{10}{1} = \frac{64}{3} = 21\frac{1}{3}$$

The Ferris wheel revolves $21\frac{1}{3}$ times in 10 minutes.

- ③ Carla baked an apple pie. She ate $\frac{1}{10}$ of it and each of her 6 friends ate $\frac{1}{8}$ of it. How much of the pie remains?

$$\frac{1}{1} - \frac{1}{10} - \frac{6}{8} = \frac{9}{10} - \frac{6}{8} = \frac{36-30}{40} = \frac{6}{40}$$

$\frac{3}{20}$ of the pie remains.

- ④ Michael completed 4 chores in 2 h. One of the chores took him $\frac{3}{5}$ h and two of the other chores each took him $\frac{2}{3}$ h. How long did the remaining chore take him?

$$\frac{2}{3} + \frac{2}{3} + \frac{3}{5} + x = 2 \Rightarrow \frac{30-29}{15} = \frac{1}{15}$$

The remaining chore took him $\frac{1}{15}$ h.

- ⑤ A jug has $3\frac{1}{3}$ L of water after $\frac{1}{6}$ of it was poured out. How much water was poured out in litres?

$$3\frac{1}{3} \times \frac{1}{6} = \frac{10}{3} \times \frac{1}{6} = \frac{5}{9}$$

$$\frac{10}{3} + \frac{5}{9} = \frac{60+10}{18} = \frac{70}{18} = 3\frac{16}{18} = 3\frac{8}{9}$$

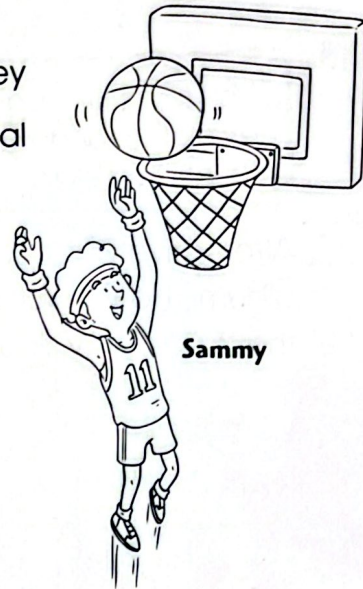
$3\frac{8}{9}$ of water was poured out.

- ⑥ Sammy and Diana were playing basketball. They each made 20 throws. Sammy scored $\frac{2}{5}$ of his total throws and Diana scored $\frac{3}{10}$ of hers.

How many more throws did I score than Diana?

$$\frac{8}{20} - \frac{6}{20} = \frac{2}{20}$$

Sammy scored 2 more throws.



- ⑦ Melanie put $1\frac{1}{2}$ bags of peanuts into a jar. She then used $\frac{1}{6}$ of the peanuts in the jar. If each bag had $\frac{4}{5}$ kg of peanuts, how many kilograms of peanuts are left in the jar?

Peanuts: $1\frac{1}{2}$ kg

$$1\frac{1}{2} - \frac{1}{6} = 1\text{ kg}$$

1 kg of peanuts are left.

- ⑧ Arnold lives $8\frac{1}{2}$ km away from a theatre. To get to the theatre, Arnold walks $\frac{1}{10}$ of the distance to a bus stop and then travels by bus.

- a. If the bus makes a stop every $\frac{17}{20}$ km, how many stops will the bus make after Arnold gets on it?

$$8\frac{4}{5} \times \frac{20}{1} = \frac{84 \times 20}{5} = \frac{84 \times 4}{1} = 336$$

$$336 - \frac{1}{10} = \frac{3360}{10} - \frac{1}{10} = \frac{3359}{10} = 335\frac{9}{10}$$

The bus will make 24 stops.

- b. There is a library at the 4th bus stop. How far is the library from Arnold's house?

$$4 \times \frac{17}{20} = \frac{32}{5} + \frac{1}{10} = \frac{34}{10} + \frac{1}{10} = \frac{35}{10} = 3\frac{1}{2}$$

The library is $3\frac{1}{2}$ km from Arnold's house.

⑨

My favourite band has released a new album available for download. Each track costs \$0.99 and there are 12 tracks in all.



Alternatively, it costs \$10.99 to download the entire album. How much more does it cost to download all the tracks individually?

Entire: 10.99
Track: 11.88

$$\begin{array}{r} 11.88 \\ - 10.99 \\ \hline 0.89 \end{array}$$

It will cost 0.89 more. ✓

- ⑩ A medical test has an accuracy rate of 0.95. If 60 tests are done, how many test results are inaccurate?

??

- ⑪ Justin drinks 0.25 L of milk every day. He plans to increase his consumption by 0.1 L each day. How many days will it take Justin to finish 4.2 L of milk?

$$0.25 + 0.35 + 0.45 + 0.55 + 0.65 + 0.75 + 0.85 + 0.95 = 4.8$$

It will take him 8 days ✓

- ⑫ Almonds are sold by weight. During a sale event, the cost of 1 kg of almonds was decreased from \$17.60 to \$12.35. Mr. Sun bought 2.8 kg of almonds. How much did he save?

NO Sale: 49.28\$
Sale: 34.58

$$\begin{array}{r} 49.28 \\ - 34.58 \\ \hline 14.70 \end{array}$$

He Saved \$14.70 ✓

- ⑬ Timothy ate 0.1 of 2.68 kg of snacks. He then divided the remaining snacks equally into 4 bags. How many kilograms of snacks were there in each bag?

$$\frac{10}{1000} \times \frac{2.68}{1} = \frac{26.8}{1000} = 0.0268$$

$$\begin{array}{r} 2.6800 \\ - 0.0268 \\ \hline 2.6532 \end{array}$$

$$2.6532 \div 4 = 0.6633$$

0.603

There are 0.6133 kg in each bag.

- ⑭ There are 5 rolls of ribbon. 4 of the rolls are each 5.5 m long and the remaining roll is 1.25 m longer. What is the total length of the ribbon?

4 roll: 22 m

Remaining roll: 23.25 m

$$\begin{array}{r} 22.00 \\ + 23.25 \\ \hline 45.25 \end{array}$$

The total length is 45.25 m

- ⑮ A 0.75-L bottle of shampoo costs \$5.25. Its value size has 0.45 L more shampoo and it costs \$2.19 more than the regular size.

a. How much does 1 L of shampoo in value size cost?

$$1.20 \text{ L} = 7.44\$$$

$$1 \text{ L} = 7.44 \div 1.2$$

$$1 \text{ L} = 6.20\$$$

1 L of Shampoo costs 6.20\$ in value size

b.

I want to buy 6 L of shampoo. How much will I save if I buy in value size?

Value: 37.2\$

Normal: 42.00\$

$$\begin{array}{r} 42.00 \\ - 37.20 \\ \hline 4.80 \end{array}$$

You will save 4.80\$



- ⑩ It cost Janet \$51.78 to fill $\frac{3}{4}$ of her car's gas tank. How much would it have cost to fill up the gas tank?

$$51.78 \times \frac{4}{3} = 51.78 \div \frac{3}{4} = 12.73 \quad \begin{array}{r} 51.78 \\ \times 12.73 \\ \hline 64.51 \end{array}$$

64.51\$ to fill up the entire tank

- ⑪ Louis is trying to sell the remaining pizzas at the end of the day. He is selling $\frac{2}{3}$ of a cheese pizza for \$5.50 and $\frac{5}{8}$ of a pepperoni pizza for \$4.55. Which pizza is a better deal?

$$\frac{2}{3} : \frac{5}{8} = \frac{16}{24} : \frac{15}{24}$$

The pepperoni is a better deal

- ⑫ The regular cost of 1 kg of beans is \$6.33. Sandra has a coupon that reduces the cost by half. How much will $\frac{1}{4}$ kg of beans cost?

$$\frac{1.588}{2} = \frac{1}{4} \text{ kg} \rightarrow \frac{1}{4} \text{ kg} = 0.79$$

It will cost 0.79\$

- ⑬ Keith paid 3 quarters to park his car for $\frac{1}{4}$ h. How much would it have cost to park the car for $1\frac{1}{2}$ h?

It would have cost 18 quarters

- ⑭ I mixed $\frac{7}{8}$ of this bottle of soda with $\frac{3}{5}$ of this carton of juice to make fruit punch.

How much fruit punch did Jason make?

$$2.31 + 0.792$$

$$= 3.102 \text{ L of}$$

He made 3.102 L of fruit punch



- ⑭ Mr. Arnold is making concrete by mixing 1.15 kg of water with $2\frac{4}{5}$ kg of cement mix. The concrete is enough to fill $1\frac{2}{3}$ holes. How much concrete is there in 1 hole?

$$\frac{1\frac{2}{3}}{1\frac{2}{3}} = \frac{3.95}{1.66} \rightarrow 1 = 2.37$$

There is 2.37 kg of concrete ✓

- ⑮ 0.84 L of tomato sauce fills $1\frac{2}{5}$ containers. How much tomato sauce should Dino make to fill $2\frac{4}{5}$ containers?

he needs 1.68 L to fill $2\frac{4}{5}$ containers ✓

- ⑯ Sandy and Connie split the cost of their meal. Sandy paid \$12.75, which was $\frac{1}{3}$ of the bill. Connie paid the remaining with a \$50 bill. What was Connie's change?

$$\text{Bill} = 38.25$$

$$\frac{2}{3} \text{ bill} = 25.50$$

Connie's change is 24.50\$ ✓

- ⑰ $\frac{1}{5}$ of a rope was 8.25 m long. Albert used 2.5 m of the rope. What is the mass of the remaining rope if each metre weighs 0.4 kg?

$$\frac{5}{5} = 41.25 \text{ m} \rightarrow \frac{5}{5} = 38.75 \times 0.4 \rightarrow 15.5 \text{ kg}$$

The mass is 15.5 kg ✓

- ⑱ The cost of an apple is $\frac{3}{4}$ of the cost of a pear. If a pear costs \$2.40, how many apples can be bought with the money needed for 3 pears?

$$3 \text{ pear} = 7.20$$$

$= \frac{28.8}{4} = 7.2$ you can buy 7 apples ✓